

ECG Vital Monitoring Task Description

Objective

Develop an algorithm to extract vital signs (ECG, SpO₂, BP, etc.) from a video of an ECG monitor using image processing and machine learning (ML) techniques. The solution should identify and extract the Region of Interest (ROI) from the monitor display and accurately report the values of the vital signs, ensuring robustness across variations in monitor designs and environmental conditions.

Additionally, you are required to propose a **stable solution** for constant interpretation of a video feed, like extracting frames in real-time while preserving image quality. Your algorithm should ensure accurate, consistent extraction without degradation of data integrity.

Tech Stack: Python, OpenCV, TensorFlow/Keras, PyTorch, or any other framework of your preference.

Key Requirements

1. **Vital Sign Extraction:**
 - Accurately extract and interpret ECG waveforms, SpO₂, BP, and other vital signs from the monitor display.
 2. **Stable Frame-by-Frame Interpretation:**
 - Propose and implement a robust method for processing video feeds frame by frame.
 - Ensure no loss in image quality during extraction, enabling consistent interpretation of data.
 3. **Handling Variations:**
 - Ensure robustness across variations in video quality, lighting conditions, and monitor layouts.
 - Handle diverse monitor types and different placements of parameters on the screen.
 4. **Dataset:**
 - Use the provided video dataset: [ECG Monitor Video Link](https://drive.google.com/file/d/1oWtAqPRQyhXaGhGSBBnwsdbydeo9wkfl/view?usp=sharing).
<https://drive.google.com/file/d/1oWtAqPRQyhXaGhGSBBnwsdbydeo9wkfl/view?usp=sharing>
 - **Important:** This video contains medically sensitive information and is shared strictly for academic purposes related to this assessment. You are not required to share it with anyone or use it outside the scope of this assignment.
 - You may supplement your work with other publicly available datasets or video resources. If you do so, clearly document the source and include this information in your final report.
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Deliverables

1. **Python Scripts:**

- Provide scripts for preprocessing, ROI detection, signal extraction, and data interpretation.
- Include a method for stable frame-by-frame interpretation of the video feed.

2. **Model Deployment Video** (Mandatory):

- Include a screen recording demonstrating your deployed algorithm extracting and interpreting vital signs from the video feed. Highlight how your solution processes the video feed in real-time while preserving image quality.

3. **Documentation:**

- Submit a well-documented PDF report or an image summary including:
 - Final accuracy scores and performance metrics for the extracted parameters.
 - Observations about the algorithm's strengths and limitations.
 - Proposed improvements and detailed explanation of the stable video feed interpretation methodology.

4. **Trained Model/Algorithm:**

- Submit any trained model files or the complete algorithm implementation.
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Evaluation

The primary evaluation criterion is **accuracy and robustness** of the extracted vital signs and the effectiveness of your frame-by-frame video feed interpretation. Proper documentation, creative approaches, and the ability to handle variations will be highly valued.

Additional Notes

- Use the provided video dataset for evaluation. Supplementary data, if used, must be properly documented, and sources must be shared.
 - Ensure that your solution is optimized for generalization across diverse scenarios (e.g., different monitor designs, lighting conditions).
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We appreciate your participation and look forward to your innovative solutions! Please reach out if you have any questions or need clarifications.

Good luck!